

PHYS 8602

Problem #2 Due 2/24/25

Consider a cubic box of fixed volume V with periodic boundary conditions. The cubic box contains $N = 256$ spherical particles which interact via a Lennard-Jones potential suitable for Argon atoms with: $\sigma = 0.3405 \text{ nm}$, $\epsilon/k_B = 119.8 \text{ K}$, $m = 6.63382 \times 10^{-26} \text{ kg}$. (Note, in reduced units $T^* = k_B T / \epsilon$, $\rho^* = \rho \sigma^3$, $E^* = E / \epsilon$.) Using a simple Verlet algorithm with a cutoff distance of $r = 2.5\sigma$, carry out an NVE Molecular Dynamics simulation for this Argon system. Choose a density $\rho^* = 0.636$ and a total (reduced) energy $E^* = 101.79$. Please answer the following questions:

- What is an appropriate time step to use for this simulation? Why?
- What is the time dependence of the kinetic energy (in reduced units)?
- What is the time dependence of the potential energy (in reduced units)?
- What is the average temperature T^* during the course of the simulation after the system has reached equilibrium?

Please remember to give error bars on your results.

You are to submit a brief report that includes results and analysis. Please make your report concise but describe how you analyzed your data and how you reached your conclusions. Include an annotated copy of your program and a copy of a summary.log.

Your report should be submitted as a single pdf file to dlandau@uga.edu. The filename should be "your_last name"_PHYS8602_prob2.pdf. Please remember to put your name on your report!